

Shahab Rahimirad

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EDUCATION

- Purdue University** Jan. 2023 - Jan 2028 (expected)
PhD Computer Science (3.75/4.0 GPA)
• **Coursework:** Natural Language Processing, Reinforcement Learning, LLM Reasoning, Computer Vision
- Sharif University of Technology** Sep. 2016 - Aug 2021
B.S. Computer Engineering (3.84/4.0 GPA)
• **Coursework:** Machine Learning, Information Retrieval, Bioinformatics, Deep Learning

PUBLICATIONS

- Bayesian Social Deduction with Graph-Informed Language Models**
S. Rahimirad, G. Gergerli, L. Romero, A. Qian, M. Olson, S. Stepputtis, J. Campbell. (*Under Review*). 2025

RESEARCH EXPERIENCE

- Purdue, Collaborative AI for Machines and People (CAMP) Lab** Aug. 2024 – Present
PhD researcher (PI: Joseph Campbell)
• LLM agents for multi-agent environments, RL Reasoning, GNNs
- Sharif University, Bachelor's Thesis** Jan. 2021 – Aug. 2021
Independent Research (PI: Hamid Beigy)
• Title: Deep learning Semantic and Syntactic Tagger for Persian Language
• Evaluated and compared RNN architectures (LSTM and GRU) and Transformers, improving over prior benchmarks across diverse corpora spanning formal, informal, and mixed-language domains

EXPERIENCE

- NuraFin Tech, Tehran, Iran** Mar. 2021 – Dec. 2022
Data Scientist
• Engineered a sophisticated Copy Trading bot that enabled real-time duplication of complex trading strategies, significantly enhancing client investment outcomes by streamlining their trading processes.
• Designed an ensemble method of XGBoost, Random Forest, and LSTM models to generate long and short signals, achieved 100% return from July 2021 to June 2022 in a volatile market
• Led a focused research team in developing ML algorithms and time-series forecasting for price analysis.
• Spearheaded the team's testing to achieve an 98% code coverage, significantly enhancing reliability.

TEACHING EXPERIENCE

- Purdue University Computer Science** Jan. 2023 – Present
Teaching Assistant
• Courses: Data Engineering, Data Science, Web Information Search, Artificial Intelligence Basics
- Sharif University Computer Engineering** Jan. 2019 – May 2021
Teaching Assistant
• Courses: Python Programming, Statistics, Embedded Systems

Other Projects

- Multi-Agent Theory-of-Mind** Apr 2025 - Present
Research Project, CAMP Lab
• Designing RL agents in the Overcooked game, modeling agent intentions with Variational Inference
- Concept Erasure in Personalized T2I Diffusion** Sep. 2024 - Dec. 2024
CS593 Computer Vision
• Implemented Erased Stable Diffusion and Evaluated the effects of BLIP personalization on erased concepts.

SKILLS

Programming Languages Python, Java, SQL, C++, C, Matlab, R, JavaScript, Go
Frameworks/Libraries Pandas, NumPy, TorchRL, TensorFlow, PyTorch, Wandb, Sklearn, Huggingface